

SPOTLIGHT TRACKING WITH AUTOMATED RECOGNITION (S.T.A.R.)

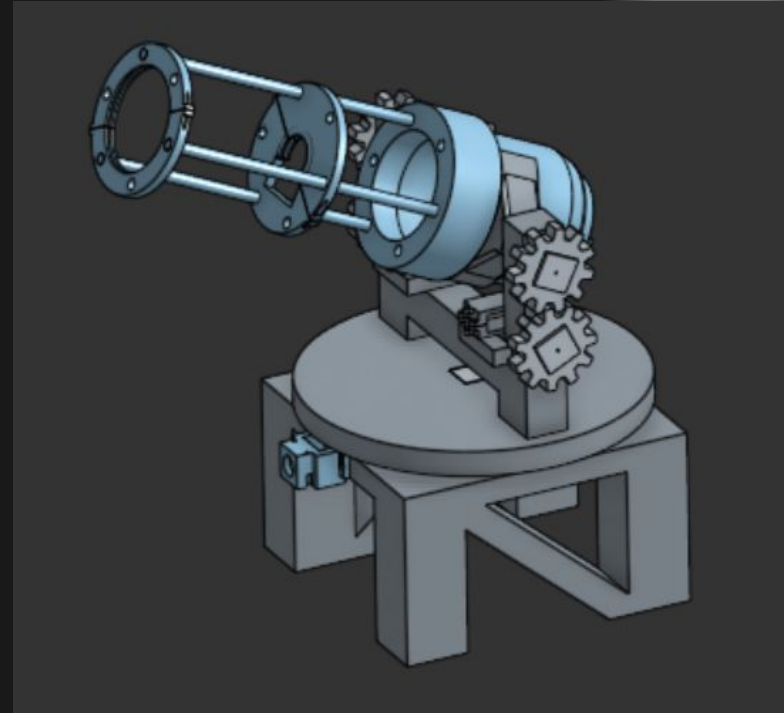


SENIOR DESIGN
SHOWCASE

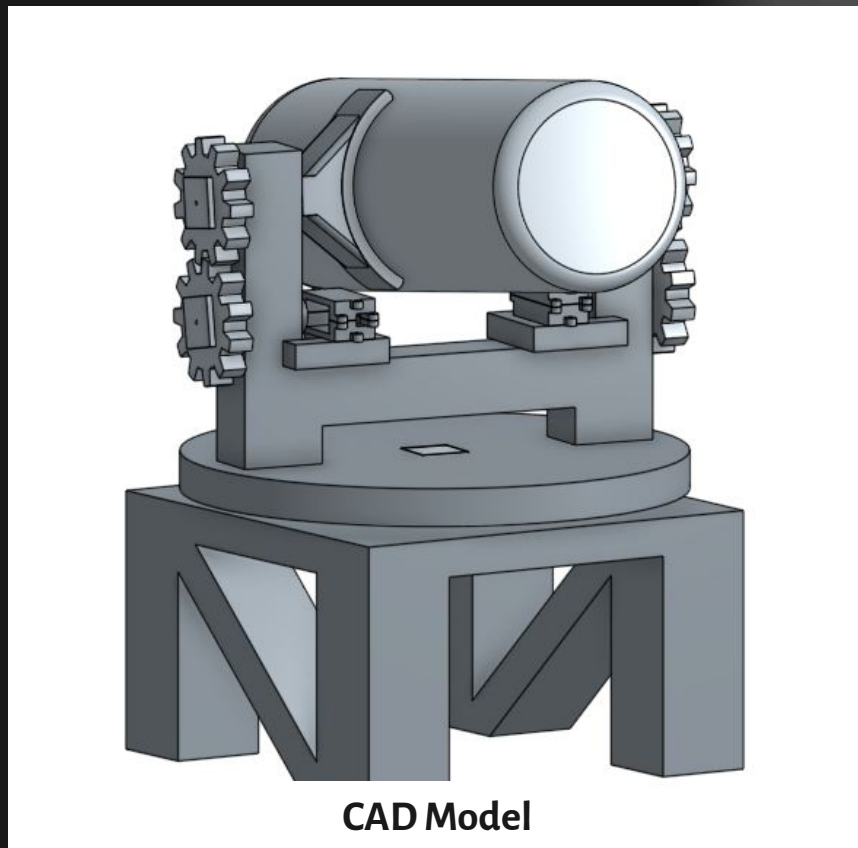


What is it?

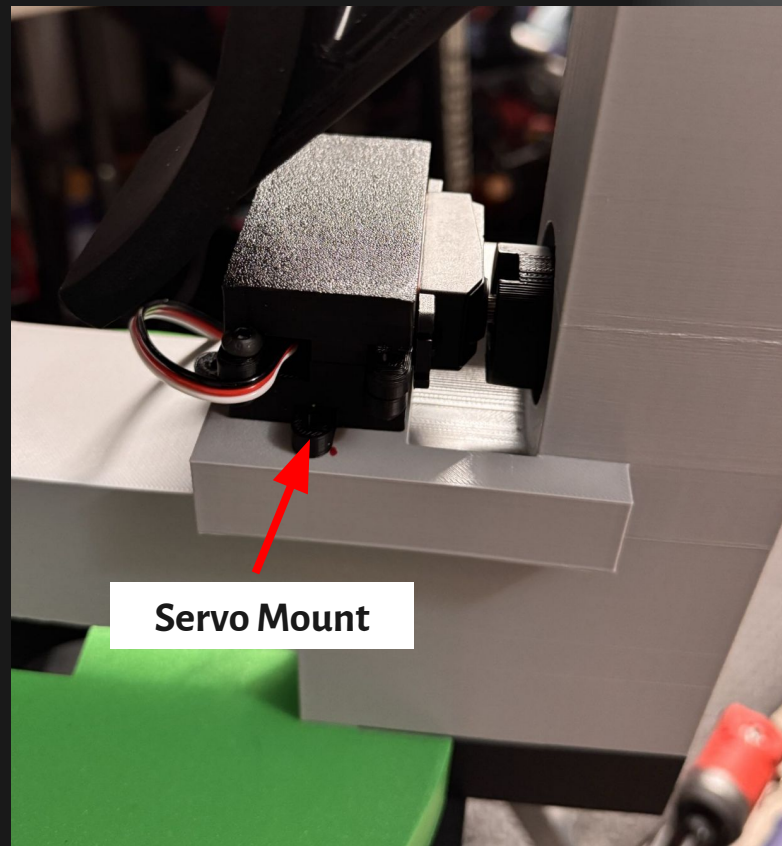
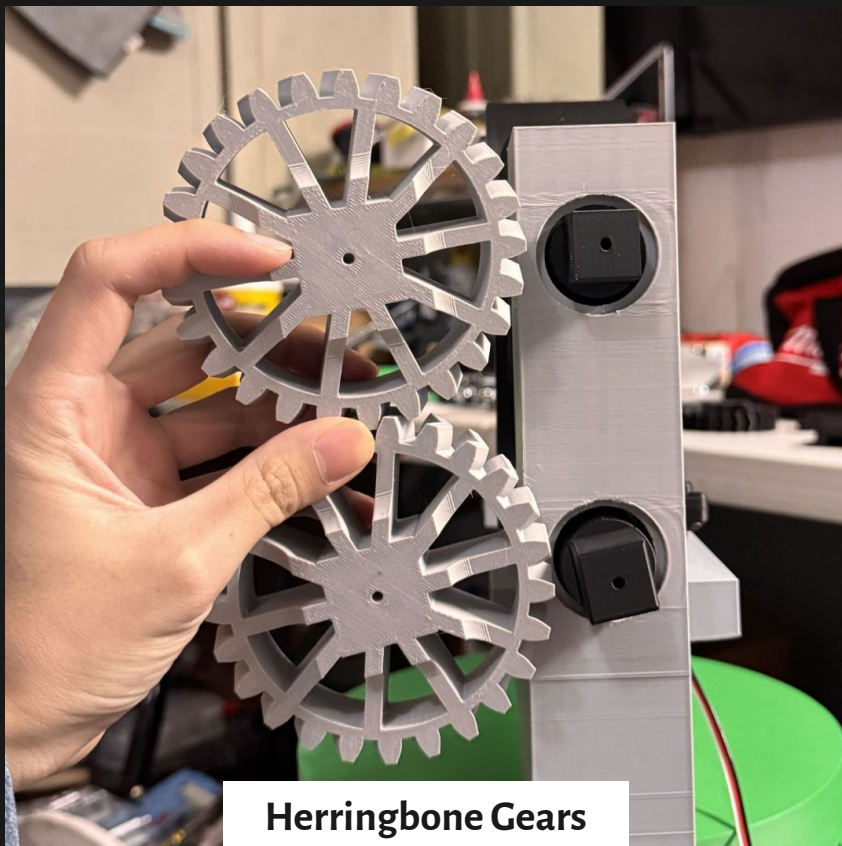
- S.T.A.R is a motorized theatrical spotlight, that can move in the x and y direction
- A camera is attached to implement human tracking, allowing a subject on stage to be automatically followed by the spotlight
- The system combines computer vision and motorized control to continuously align the spotlight with the detected performer.



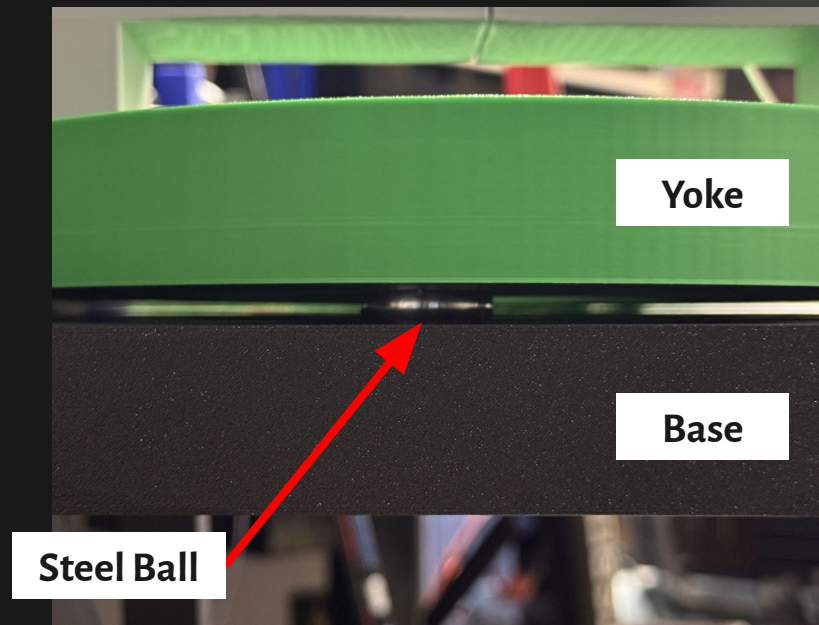
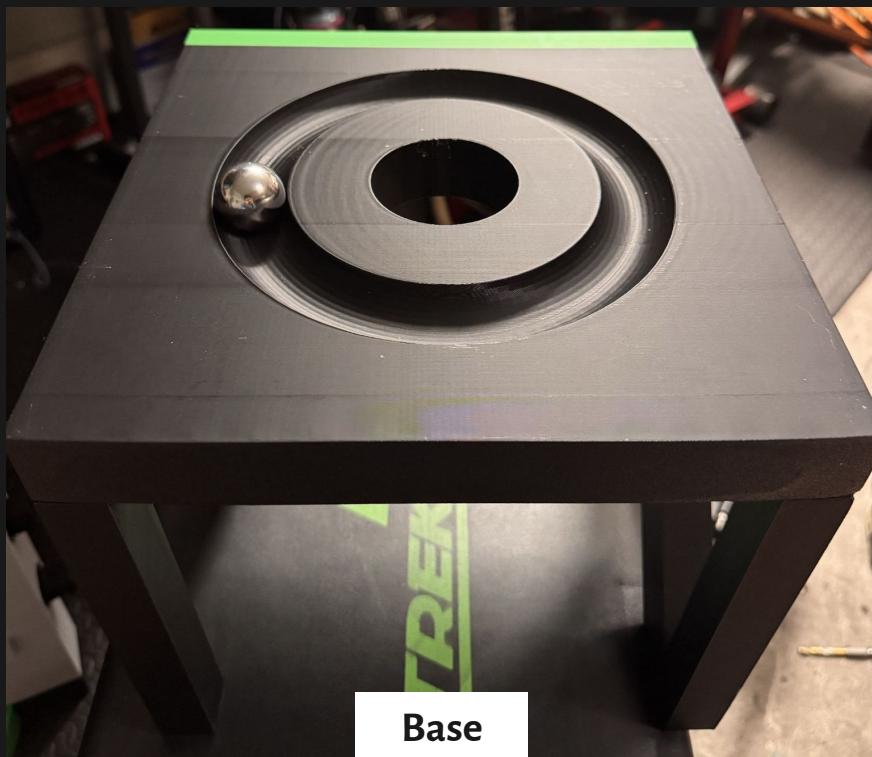
Gimbal



Gimbal



Gimbal



Key Specifications

Parameter	Description	Target Value & Unit(s)
Tracking Latency	Time delay between performer movement and spotlight response	≤ 500 ms (basic), ≤ 300 ms (advanced)
Tracking Accuracy	Angular error between spotlight center and the selected performer's center of mass	$\leq 5^\circ$ RMS at 10m
Detection Confidence	The confidence score given by the detection system upon first detection of an incident subject	$>70\%$
Field of View (FOV)	The angular viewable area that the sensor can image	$> 45^\circ$
Depth of Focus	The acceptable focus depth that a performer can move on a stage	> 1 m
Tracking Range	Distance at which subject can be reliably detected and tracked	7.5m (± 1 m)
Frames Per Second (fps)	How many frames are captured by the camera per second	30 fps @ 1920x1080p
Lighting Conditions	Minimum illumination where tracking is possible	≥ 1000 Lumens

TESTING STAGE

Illumination System Testing - Results

Center	
Degree	Lux
0	571.9

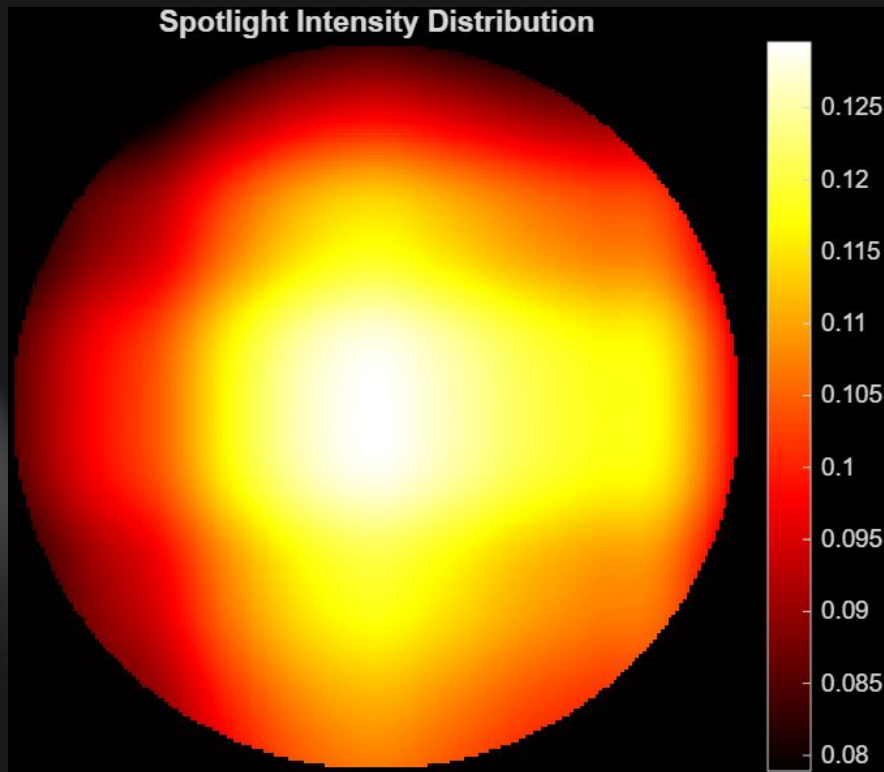
Ring 3	
Degree	Lux
0	323.6
45	326.4
90	338.2
135	344
180	344.4
225	309
270	312.8
315	281.9
360	323.6

Radius: 840mm

Spot Uniformity	
Degree	Efficiency
0	57%
45	57%
90	59%
135	60%
180	60%
225	54%
270	55%
315	49%
360	57%

- By dividing the average outer edge lux (Ring 3) by the average center lux we can find the uniformity efficiency along each degree
- Overall, the edge intensity meets our benchmark of 50% of the center

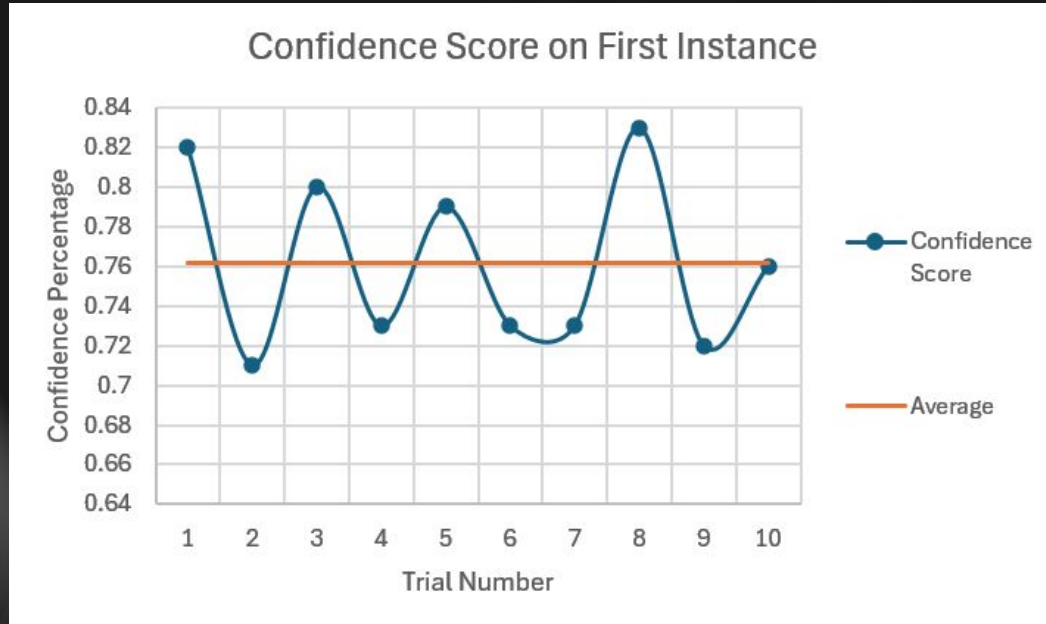
Intensity Heatmap



Overall good uniform with slower decay on the top and left edges.

Bright spot is in the center of the output, allowing focus to be drawn to the subjects face and upper body.

Tabulated Results



Trial	Detected	Confidence
1	✓	0.82
2	✓	0.71
3	✓	0.8
4	✓	0.73
5	✓	0.79
6	✓	0.73
7	✓	0.73
8	✓	0.83
9	✓	0.72
10	✓	0.76

The detection system held up very well during testing. We had an average confidence of 77% through all 10 trials.

Tabulated Results

Trial ▼	Change in Frame ▼	Latency (ms) ▼
1	92	383.64
2	120	500.4
3	73	304.41
4	83	346.11
5	99	412.83
6	86	358.62
7	66	275.22
8	111	462.87
9	123	512.91
10	97	404.49

Multiplying each change in frame by 4.17ms results in our latency.

As expected we do receive some latency from human motion to gimbal movement with an average latency of 396.2ms

$$\frac{1}{240fps} = 4.17ms$$

Each frame is 4.17ms

Trial 7 Calculations

$$\Delta Frame = 5982 - 5916 = 66$$

There is a difference of
66 frames

$$66 \text{ frame} * \frac{4.17 \text{ ms}}{\text{frame}} = 275.22 \text{ ms}$$

The latency is 275.22ms!

THE END



Thank you for watching!

THE END

THANK YOU FOR WATCHING



GROUP 7
S.T.A.R.

